

KUIYI (CORY) GAO

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PROFILE

Computer science undergraduate with research experience in agent evaluation, runtime safety, and multimodal AI. I build Python evaluation pipelines, browser tasks, automated judges, and runtime checks, and I have carried projects from method design through experiments, figures, and paper writing.

Languages: Mandarin (native), English (TOEFL iBT 110), Cantonese (intermediate).

EDUCATION

University of North Carolina at Chapel Hill Jan 2026–May 2027
B.A. in Computer Science, GPA 3.68/4.00 Chapel Hill, NC

Coursework. Algorithms, machine learning, natural language processing, technology ethics, C/Unix systems, and object-oriented design.

The Chinese University of Hong Kong Sep 2023–May 2026
Computer Science, Faculty of Engineering ELITE Stream Hong Kong

Coursework. Graduate deep learning, artificial intelligence, data structures, formal languages, computer organization, probability, statistics, and linear algebra.

RESEARCH EXPERIENCE

RealGUINoise: GUI-Agent Robustness under Interface Noise [1] Sep 2025–present
Co-first author; CUHK, advised by Prof. Michael R. Lyu Hong Kong / remote

Engineering. Built paired web and Android tasks and browser automation for controlled overlays, dynamic content, layout shifts, and look-alike controls.

Evaluation. Implemented automated judging and paired measures for task completion, unsafe actions, and action-sequence changes.

RuntimeTickets: Runtime Enforcement for Third-Party Agent Skills [2] [\[code\]](#) Jun 2026–present
First author; MBZUAI UGRIP, advised by Dr. Steve Liu Abu Dhabi / Beijing

Method. Designed task-scoped contracts and runtime gates that narrow host permissions and check covered operations before execution.

Engineering. Built decision and effect logging in a four-student team; lead method development, experiments, and manuscript writing.

Chain-of-Jailbreak: Multi-Turn Safety Attacks on Image Generation [4] Jun 2024–Feb 2025
Second author; Tencent AI Lab and CUHK Hong Kong

Research. Designed the multi-turn image-editing attack and prompting defense, contributed benchmark data, and ran most experiments.

Delivery. Wrote evaluation scripts and prepared figures, manuscript text, and rebuttal.

MMSafeAware: Benchmarking Multimodal Safety Awareness [3] Dec 2023–Oct 2024
Third author; Tencent AI Lab Hong Kong / Shenzhen

Benchmark. Constructed data and evaluation for 1,500 image-text pairs across 29 safety scenarios.

Model work. Ran vision-centric fine-tuning, analyzed missed risks and benign false alarms, and contributed to submission and rebuttal.

Text-KEN: Comparing Collections of Text May–Aug 2025
Summer research with Prof. Farzan Farnia, CUHK Hong Kong

Engineering. Built a Python pipeline that compares text collections using sentence embeddings and distribution-level statistics, without training a classifier.

PUBLICATIONS AND MANUSCRIPTS

[1] **Kuiyi Gao*** et al. (2026). *RealGUINoise: A Dynamic Cross-Platform Benchmark for GUI Agent Robustness under Real-World Interface Noise*. **Co-first author**. Under review.

[2] **Kuiyi Gao**. (2026). *RuntimeTickets: Source-Grounded Security Obligations for Agent Skills*. **First author**. Under review at ICLR 2027.

[3] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. (2025). *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 Main. [\[paper\]](#) [\[code\]](#)

[4] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. (2025). *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [\[paper\]](#) [\[code\]](#)

SELECTED PROJECTS

MemSlot: Contract Compression before OCR and Question Answering [\[code\]](#) UNC, 2026
Purpose. Tests whether learned saliency can shorten rendered contracts without discarding answer-bearing content.
Work. Built memory-slot attention, overlap-aware scoring, saliency-guided rendering, and pruning baselines; end-to-end OCR-to-QA evaluation remains incomplete.
Output. Training and evaluation notebooks, pruning comparisons, and saved word-saliency visualizations.

Poetry and Image Variation: Comparing Poetic Movements CUHK, 2025
Purpose. Studies whether Imagist and Romantic excerpts produce different visual variation in generated images.
Work. Generated images with FLUX and Stable Diffusion and compared saved CLIP-based scores in Python.
Output. Generated-image grids and plots comparing the saved variation scores across poetry groups.

Movability Segmentation: Four-Class Scene Understanding [\[code\]](#) CUHK, 2025
Purpose. Maps scene elements into movability classes for action-oriented perception.
Work. Adapted DeepLabV3 for four-class segmentation and implemented data remapping, preprocessing, training, and checkpoint inference.
Output. Course report, training code, saved checkpoints, and prediction overlays on archived scenes.

SELECTED DISTINCTIONS

Professor Joseph Lau Scholarship of Innovation
CUHK Faculty of Engineering ELITE Stream Award
CUHK College Entrance Scholarship
CUHK Exchange Scholarship for UNC Spring 2026
Gaokao, top 2% in Beijing 2023

TECHNICAL SKILLS

AI and ML engineering. Python, PyTorch, Transformers, embeddings, fine-tuning, and semantic segmentation.
Agent systems and evaluation. Browser automation, tool integration, automated scoring, runtime policies, and trace logging.
Software engineering. Git, Linux, Docker, Playwright, HTTP/JSON APIs, testing, and experiment configuration.
Analysis and communication. Statistics, data visualization, technical writing, research figures, and experiment documentation.